

Optimization of Stator Core Shape to Minimize Harmonic Current in Surface Permanent Magnet Synchronous Motor Using Topology Optimization Algorithm

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Abstract— One of the causes of torque ripple in a motor is the harmonic current. To minimize the harmonic current, we focused on minimizing the inductance harmonics in this study. The stator core shape that reduced the inductance harmonics was identified using a topology optimization algorithm. A subsequent sensitivity analysis of the stator core shape with respect to the inductance harmonics yielded preliminary evaluation shapes. Experimental assessment of the evaluation shapes verified the reduction of the harmonic current and, therefore, the torque ripple.

Keywords— Topology optimization, Inductance harmonics, Harmonic current, Torque ripple

I. INTRODUCTION

As the movement to reduce environmental loads progresses, the number of electric drive systems represented by motors is increasing. Increasing the penetration rate of motors in the human environment also increases the requirements for motor specifications such as low vibration, low noise, small size, light weight, and high efficiency. Therefore, this study focuses on improving the motor magnetic circuit with the aim of achieving low vibration and low noise of the motor.

The vibration and noise of a motor are caused by the electromagnetic excitation force in the motor. The electromagnetic excitation force of a radial gap motor can be roughly classified as a radial force, which is a radial component, and a torque ripple, which is a circumferential component. In this study, we consider the torque ripple that easily excites the eigenvalue of the motor mounting destination as the vibration of the motor output shaft. Torque ripple is considered to consist of four factors: the cogging torque, induced voltage harmonics, magnetic saturation, and time harmonic current [1]. It has been established that the cogging torque can be reduced by decreasing the no-load induced voltage harmonics [2].

In this study, the sixth-order torque ripple of an 8-pole 12-slot surface permanent magnet synchronous motor (SPMSM) having the specifications in Table 1 was targeted for reduction. Because the 8-pole 12-slot motor has a large harmonic winding factor of 0.866 for the fifth and seventh orders, the fifth- and seventh-order components are sufficiently large to appear in the induced voltage [3].

To reduce the sixth-order torque ripple, the fourth-, sixth-, and eighth-order inductance harmonics of the induced voltage harmonics under load and the fifth- and seventh-order

harmonic currents generated by them were targeted for reduction. The shape of the stator core to be reduced was obtained using topology optimization. Next, based on the results of the topology optimization, a sensitivity analysis was performed with respect to the reduction of inductance harmonics, and the evaluation shape of the stator core was determined. Finally, evaluation shapes were prepared, and the effect of reducing the fifth and seventh harmonic currents and the sixth torque ripple was evaluated in an actual machine.

II. TOPOLOGY OPTIMIZATION

A. Objective function

First, we examined the topology optimization objective function for reducing the inductance harmonics. The interlinkage magnetic flux of the three-phase motor is given by the following equation:

$$\begin{bmatrix} \Psi_u \\ \Psi_v \\ \Psi_w \end{bmatrix} = \begin{bmatrix} \Psi_{mu} \\ \Psi_{mv} \\ \Psi_{mw} \end{bmatrix} + \begin{bmatrix} L_{uu} & M_{uv} & M_{wu} \\ M_{uv} & L_{vv} & M_{vw} \\ M_{wu} & M_{vw} & L_{ww} \end{bmatrix} \begin{bmatrix} I_u \\ I_v \\ I_w \end{bmatrix} \quad (1)$$

where Ψ_u, Ψ_v , and Ψ_w are the three-phase flux linkages; Ψ_{mu}, Ψ_{mv} , and Ψ_{mw} are the three-phase flux linkages due to the permanent magnet; L_{uu}, L_{vv} , and L_{ww} are the three-phase self-inductances; M_{uv}, M_{vw} , and M_{wu} are the three-phase mutual inductances; and I_u, I_v , and I_w are the three-phase currents.

The second term on the right side of (1) is the flux linkage from the product of the inductance and current, and it is necessary to separate this from the first term. In the topology optimization algorithm, however, because separating the first term causes an increase in optimization time, it is difficult to set the inductance harmonics as the objective function for topology optimization, and other objective functions are used.

As a substitute objective function for the inductance harmonics, we considered the distribution of the harmonic flux densities on the stator core. The second term on the right side of (1) can be extracted using the frozen-permeability (FP) method [4]. The harmonic flux density distribution caused by the inductance harmonics was thus visualized. Fig. 1 shows

Table 1. Motor specifications

Max. Torque [Nm]	4.5
Max. Output [W]	550
Max. Phase Current [Arms]	76
Max. Line Voltage [Vp-p]	9.8

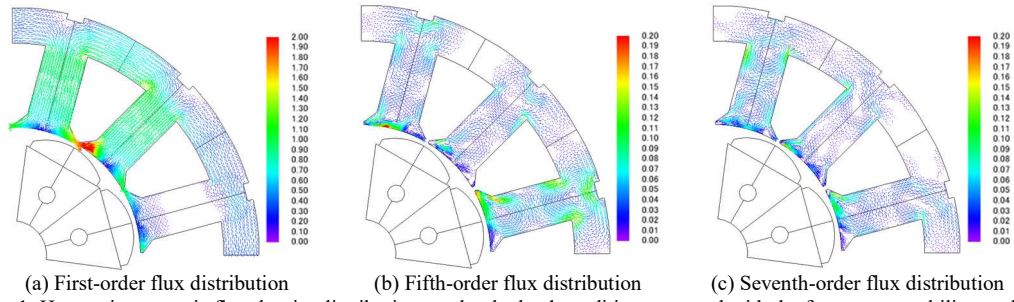


Fig. 1. Harmonic magnetic flux density distributions under the load condition extracted with the frozen-permeability method

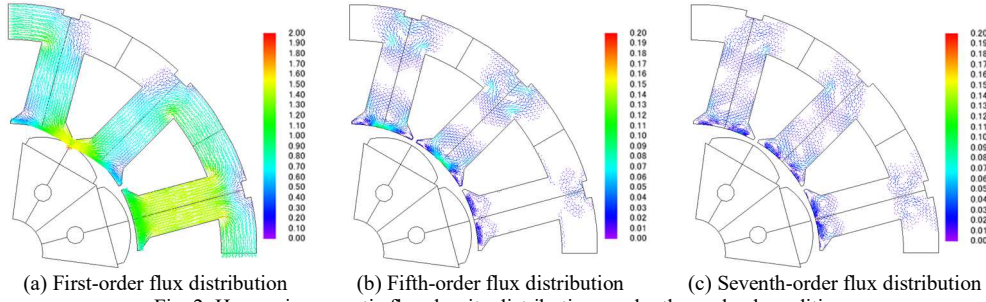


Fig. 2. Harmonic magnetic flux density distributions under the no-load condition

the magnetic flux density distributions of the first, fifth, and seventh orders on the stator core extracted using the FP method for the target motor under the load condition. Because the motor is an SPMSM, only the q-axis current contributes to generating magnetic torque, and the first-order magnetic flux flows such that it rotates around the two stator teeth that are not facing the permanent magnet. On the other hand, the fifth- and seventh-order magnetic fluxes have high magnetic flux density distributions on the stator teeth facing the permanent magnets.

Fig. 2 shows the first-, fifth-, and seventh-order magnetic flux density distributions on the stator core under the no-load condition. The first-order magnetic flux under the no-load condition has a high magnetic flux density on the stator teeth facing the permanent magnet on the rotor side. However, the magnetic flux densities of the fifth and seventh orders on the stator teeth facing the permanent magnet are low.

Therefore, as shown in Fig. 3, the magnetic flux flowing through the magnetic path generated by the mutual inductance between the v and w phases under the load condition attempts to short-circuit to the u -phase stator teeth, as shown by the red dotted lines. Then, the fifth and seventh fluxes appear. The fifth and seventh flux density components in the center of the stator facing the permanent magnet were used as the objective function for the topology optimization to identify shapes for reduction of the inductance harmonics.

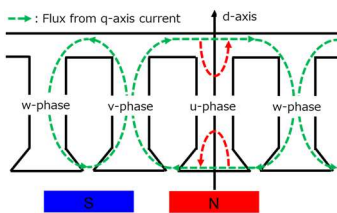


Fig. 3. Schematic diagram of causes of harmonic magnetic flux

B. Optimization conditions

Next, the conditions for topology optimization are described. Fig. 4 shows the design area for topology optimization. The design area comprises one stator tooth, and the constituent materials are two types of electrical steel parts and air. In addition, a mirror symmetry plane is provided in the center of the stator tooth, and the mesh is mirror-copied in the tangential direction. The objective function to be minimized is the sum of the amplitudes of the tangential components of the fifth- and seventh-order magnetic flux densities extracted from these six coordinates by dividing the center of the stator teeth into six regions in the radial direction.

Table 2 lists the conditions of the normalized Gaussian network (NGnet) [5]. Load conditions of 2 Nm and 1000 rpm were used for each analysis. In addition, the continuity of the shape was not included in the evaluation criteria, and the cases on the Pareto chart were evaluated.

C. Optimization results

Fig. 5 shows the distribution of the objective functions. The red dotted line represents the value of the objective function for the original shape before optimization, and it was

Table 2. Topology optimization condition

Max. Generation	50
Number of Parents	500
Number of Children	500
Number of NGnet	114

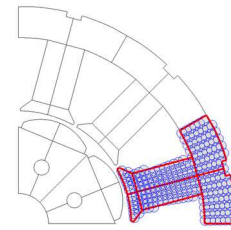


Fig. 4. Design area

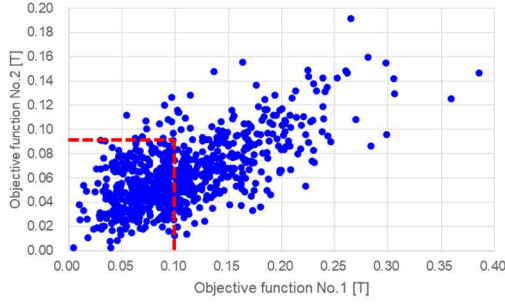


Fig. 5. Distribution of the objective functions for each shape

determined that a sufficient number of solutions were obtained. Examples of the obtained shapes are shown in Fig. 6 (a), (b), and (c), and the values of the objective functions are listed in Table 3. The mesh displayed in red represents the mesh that becomes the electrical steel parts. The shape trends can be roughly divided into three types, as listed below. The flow of the harmonic flux of the red dotted line is hindered by the blue line, as shown in Fig. 6 (d), (e), and (f).

- Shave the end of the tip of the stator tooth
- Create a hole in the center of the stator tooth
- Hollow out the base on the tip side of the stator tooth

III. EVALUATION SHAPE DESIGN

There is a problem in prototyping the mesh shapes obtained from the topology optimization because the shapes are too complex to reproduce in the actual product. Therefore, we performed a sensitivity analysis of the obtained mesh shapes with respect to reducing the inductance harmonics and the mesh with low sensitivity to the reduction of inductance harmonics was changed to electrical steel sheet in the prototype.

The mesh shapes shown in Fig. 6 (a) and (c) obtained by topology optimization were rotationally copied in the tangential direction to create evaluation shapes for the 8-pole 12-slot motor, and the inductance waveforms were evaluated. The mesh shape shown in Fig. 6 (b) was excluded because torque reduction due to magnetic saturation was likely to occur. It was assumed that the inductance is a differential inductance and the flux linkage due to the permanent magnet

Table 3. Objective functions after optimization

	Objective function No. 1 [T]	Objective function No. 2 [T]
Original shape	0.103	0.086
Example shape 1	0.017	0.049
Example shape 2	0.018	0.011
Example shape 3	0.014	0.033

does not change when only the u -phase current is slightly increased from a certain current condition, as in (2). The differential inductance was calculated based on the change in the flux linkage, as shown in (3). The same procedure was performed for the other two phases, and each inductance was calculated.

$$\begin{bmatrix} \Psi_u' \\ \Psi_v' \\ \Psi_w' \end{bmatrix} = \begin{bmatrix} \Psi_{mu} \\ \Psi_{mv} \\ \Psi_{mw} \end{bmatrix} + \begin{bmatrix} L_{uu} & M_{uv} & M_{wu} \\ M_{uv} & L_{vv} & M_{vw} \\ M_{wu} & M_{vw} & L_{ww} \end{bmatrix} \begin{bmatrix} I_u + \Delta I_u \\ I_v \\ I_w \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} \Psi_u' \\ \Psi_v' \\ \Psi_w' \end{bmatrix} - \begin{bmatrix} \Psi_u \\ \Psi_v \\ \Psi_w \end{bmatrix} = \begin{bmatrix} L_{uu} \\ M_{uv} \\ M_{wu} \end{bmatrix} \Delta I_u \quad (3)$$

Where Ψ_u' , Ψ_v' , and Ψ_w' are the three-phase flux linkages when the u -phase current is slightly increased, and ΔI_u is the amplitude of the u -phase validated current.

Fig. 7 shows a comparison of the shapes of the stator cores before and after the sensitivity analysis. After the sensitivity analysis, the shape of Fig. 7 (a) was filled with electrical steel parts, except for the scraped part at the tip of the stator teeth, as shown in Fig. 7 (b). After the sensitivity analysis, the shape of Fig. 7 (c) was filled with electrical steel parts except for the hollowing out of the base of the tip of the stator teeth, as shown in Fig. 7 (d).

Fig. 8 shows comparisons of the amplitudes of the u -phase self-inductance harmonics, u -phase harmonic current, and torque ripple for the original stator core shape and the two evaluation shapes under the finite element analysis (FEA) conditions summarized in Table 4. The u -phase harmonic current and torque ripple are obtained when the analysis is performed with the sinusoidal voltage from the analysis values of the three-phase voltage under a sinusoidal current. In Fig. 8 (a), evaluation shape 1 can reduce the fourth-, sixth-, and eighth-order components of the inductance, and evaluation shape 2 can reduce the fourth- and sixth-order components of

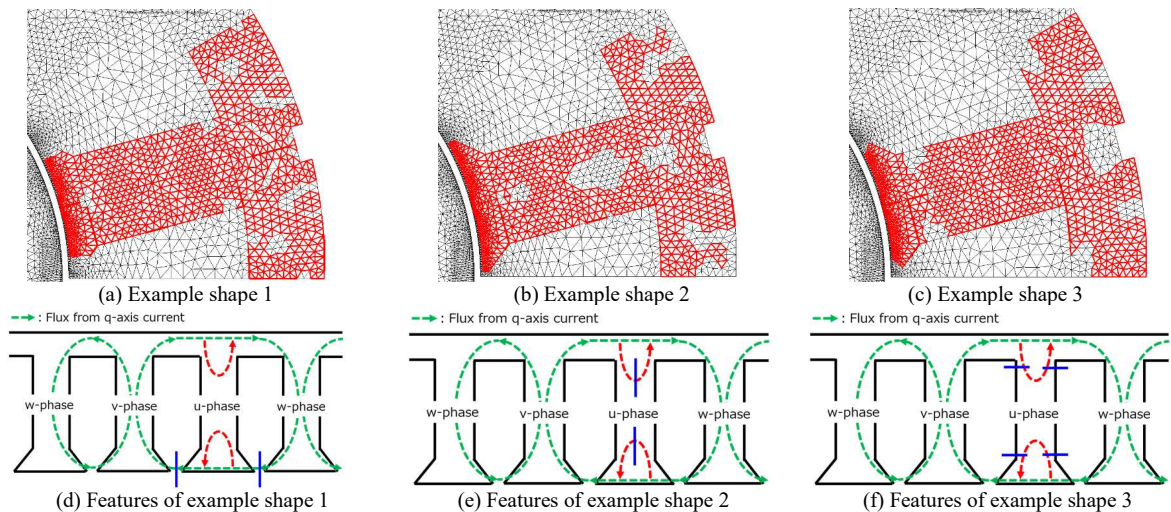


Fig. 6. Several examples of topology optimization and features of each shape

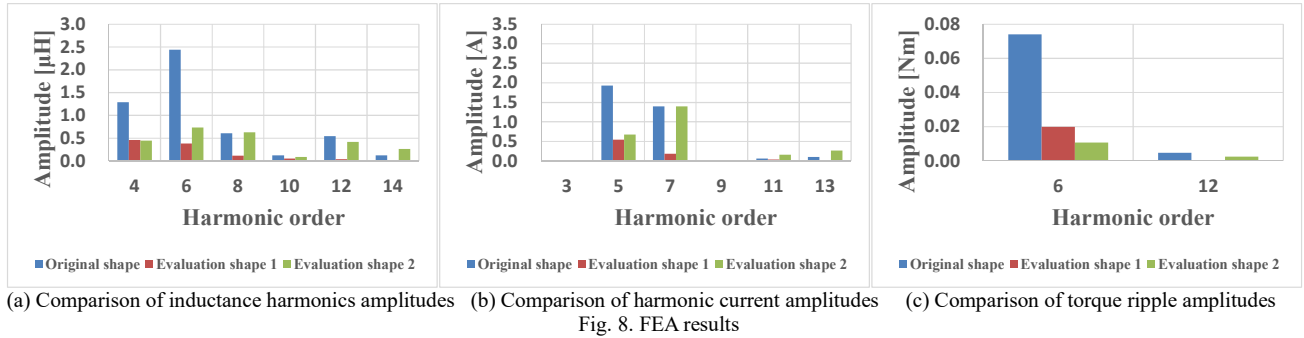
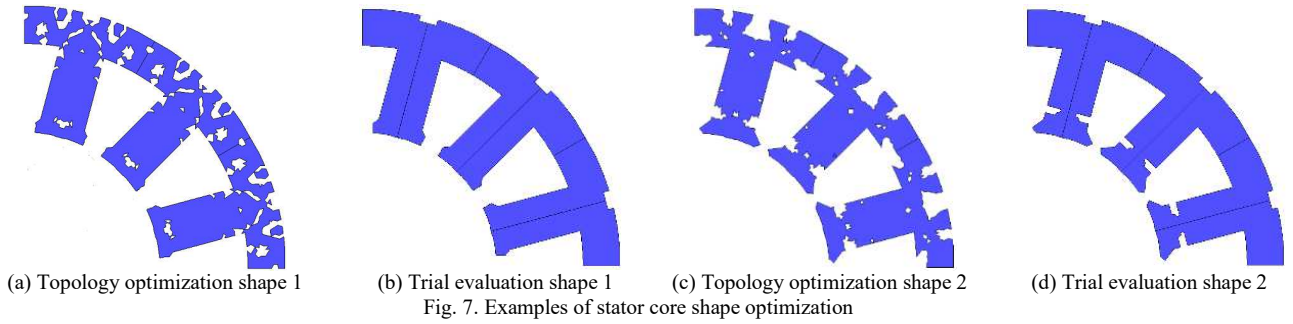


Table 4. FEA conditions

Torque [Nm]	4
Rotational speed [rpm]	1000
Fundamental phase current [Arms]	66.4
Fundamental phase angle [degrees]	0

the inductance compared with the shape before optimization. Owing to this inductance harmonic reduction effect, in Fig. 8 (b), the fifth- and seventh-order harmonic currents of evaluation shape 1 are reduced, and the fifth-order harmonic current of evaluation shape 2 is reduced. In addition, because the FEA confirmed that the sixth torque ripple could be reduced by more than half in the two evaluation shapes, the actual motor was evaluated.

IV. EXPERIMENTAL RESULTS

The experimental environment is shown in Fig. 9. A load was applied while controlling the rotation speed of the motor with a servo motor, and a current command value or voltage command value was provided by the controller to drive the motor for evaluation. A current sensor with a built-in inverter was used to evaluate the harmonic currents. A torque sensor and a fast Fourier transform (FFT) analyzer were used to evaluate the torque ripple, and an averaging process was performed 100 times using the FFT analyzer. The experimental conditions are listed in Table 5, and the control system for the controller is shown in Fig. 10. First, the same

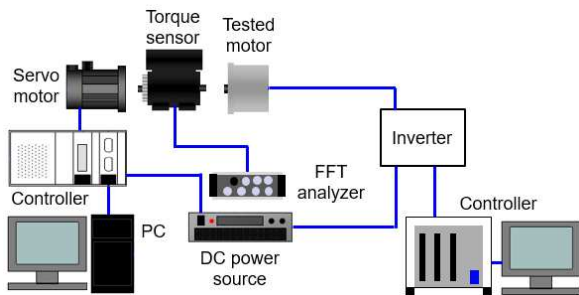


Fig. 9. Experimental environment

current command value was given to all three types of motors, and the motor was driven by the current feedback control system shown in Fig. 10 (a) to obtain a steady-state voltage command value in the controller. Next, the voltage command values obtained for each of the three types of motors were driven by the voltage control system shown in Fig. 10 (b), and the harmonic currents and torque ripple were evaluated.

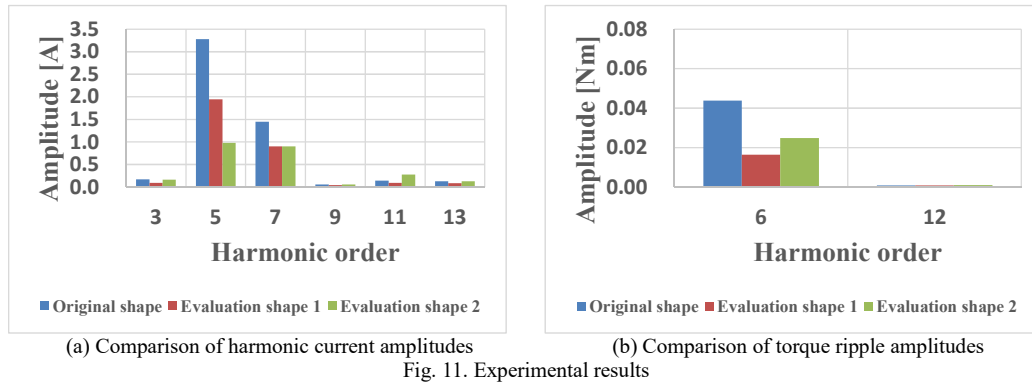
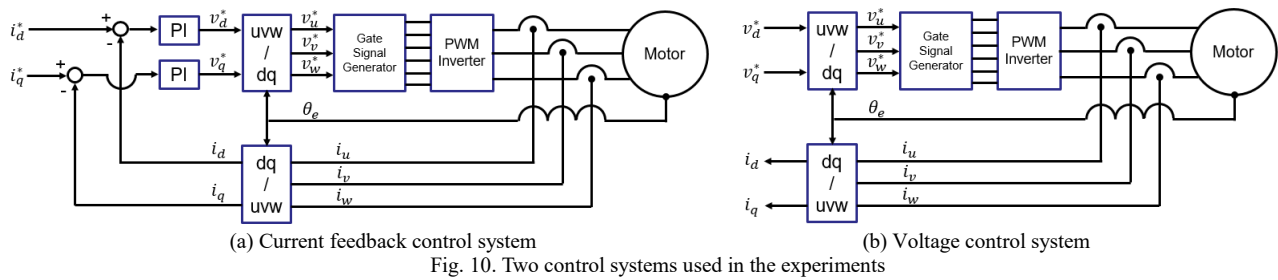
Fig. 11 shows comparisons of the amplitudes of the u -phase harmonic currents and torque ripples between the stator core shape before topology optimization and the two evaluation shapes in the actual machine. The harmonic currents of the fifth and seventh orders are reduced with both evaluation shapes 1 and 2. In addition, the sixth torque ripple is reduced by approximately half. The qualitative tendency is in agreement with the FEA results in Fig. 8, but the magnitudes of the reductions differ.

V. CONCLUSION

In this study, to reduce the sixth-order torque ripple, the fourth-, sixth-, and eighth-order inductance harmonics of the induced voltage harmonics under a load and the fifth- and seventh-order harmonic currents generated by them were targeted for reduction. The shape of the stator core to be reduced was determined using topology optimization. Based on the obtained optimization results, the evaluation shape of the stator core was determined, and the fifth and seventh harmonic currents and sixth torque ripple were evaluated using FEA and actual machine evaluations.

Table 5. Experimental conditions

Torque [Nm]	4
Rotational speed [rpm]	1000
Fundamental phase current [Arms]	66.4
Fundamental phase angle [degrees]	0
q-axis current command value [A]	115
d-axis current command value [A]	0
Career frequency [kHz]	20
Dead time [μs]	1



Both the FEA results and the actual machine evaluations showed reductions in the fifth and seventh harmonic currents and the sixth torque ripple.

In the future, we will investigate the factors behind the discrepancy between the analysis and experimental results and evaluate motor characteristics such as the cogging torque and torque characteristics.

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